MILK10k Skin-Lesion Classification — Methods, Results and Lessons Learned

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Authorship note: the work described here was largely done by Claude Opus 5.5 (Anthropic), with guidance and experience contributed by the author.

Abstract

The best model so far scores 0.584 macro F1 in 5-fold cross-validation (CV) and 0.546 on the MILK10k Benchmark leaderboard (rank 16 end of September 2026), using only the MILK10k training data. It is a PanDerm-Base vision transformer fine-tuned separately on the dermoscopic and the clinical close-up image, with the two predictions combined by a geometric mean (late fusion), averaged over 3 seeds, and per-class decision thresholds fitted on out-of-fold predictions.

The work proceeded in pre-specified steps, each compared to the previous best with a paired lesion bootstrap. Three changes produced resolved gains: image models instead of metadata (0.363 → 0.453), full fine-tuning instead of frozen features (→ 0.548–0.560), and adding the clinical image by late fusion (→ 0.584). Eight further ideas gave no gain that is resolvable at this sample size: metadata fusion, joint two-view models, a different augmentation pipeline, 448 px input, label-shift correction, classifier re-training with class balancing, other epoch budgets, and post-hoc ensembles.

On the test set the model ranks lesions well (mean AUC 0.929, above the leader's 0.905), but loses F1 on four rare classes (MAL_OTH, BEN_OTH, INF, VASC; about 33 test lesions). These account for 79% of the 0.159 gap to rank 1, which declared the use of external data. More rare-class data is therefore the most likely remaining lever — a hypothesis, not yet tested.

1 Introduction

The MILK10k Benchmark (ISIC; organisers H. Kittler and P. Tschandl, Medical University of Vienna) asks for an 11-class skin-lesion diagnosis from a pair of images per lesion — one clinical close-up and one dermoscopic image — plus metadata. The metadata are age (5-year groups), sex, skin tone (0–5), anatomical site and MONET concept scores (e.g. ulceration, vessels, pigmentation).

The 11 classes are AKIEC, BCC, BEN_OTH, BKL, DF, INF, MAL_OTH, MEL, NV, SCCKA and VASC. Submissions give 11 probabilities per lesion for 479 blind test lesions.

The metric is macro F1 ("Dice"): each class is binarised at probability ⩾ 0.5, F1 is computed per class, and the 11 values are averaged without weighting. Two consequences drove many decisions in this project:

- Every class counts 1/11, so a class with ~10 test lesions weighs as much as BCC.
- The fixed 0.5 cut-off makes the score depend on calibration and thresholds, not only on how well the model ranks lesions (AUC).

The benchmark also states that the test set contains granular diagnoses not present in training, mainly within BEN_OTH and MAL_OTH.

2 Data

Only the MILK10k training set was used: 5,240 lesions, each with one clinical and one dermoscopic image of 600×450 px. No patient ID is provided, so folds are lesion-level.

The classes are extremely imbalanced. Five classes together have 202 lesions (3.9%); MAL_OTH has only 9, all of them collision tumours.

Class	Training lesions	Test lesions (derived, ±1)
BCC	48%	61 (13%)
NV	~14%	52
BKL	~10%	100
SCCKA	~9%	117
MEL	~9%	39
AKIEC	~6%	71
DF	52	6
INF	50	11
VASC	47	5
BEN_OTH	44	7
MAL_OTH	9	~10

The test counts were back-calculated from the leaderboard's per-class sensitivity and PPV of our own submission. They were used only to explain results, never to tune a model, because that would fit the hidden labels. They show a strong prior shift: BCC drops from 48% to 13%, SCCKA, BKL and AKIEC rise.

External data. No external images were used. The only indirect external information is in the pretrained weights: PanDerm-B (>2 M dermatology images; overlap with MILK10k sources not verified), EVA-02 (ImageNet-22k) and DINOv2 (LVD-142M). A local inventory of candidate external sources was made (ISIC archive, MRA-MIDAS, Derm7pt, PAD-UFES-20, DDI, Fitzpatrick17k). It found that the ISIC archive contains the MILK10k training images themselves ("MILK study team", 5,211 lesions) and that several test-only diagnoses (Merkel cell carcinoma, atypical fibroxanthoma, cylindroma, spiradenoma, lymphangioma, AV malformation) are absent from all local sources.

3 Methods

3.1 Evaluation protocol (fixed before any modelling)

The protocol was designed so that small differences could be told apart from noise, since the leaderboard (479 lesions, ~10 per rare class) cannot do that.

- **Folds:** one canonical stratified 5-fold split (seed 42), reused by every step; every CV number is out-of-fold (OOF).
- **Metric:** the challenge rule (per-class F1 at ⩾ 0.5, macro mean).
- **Nested threshold calibration ("calib"):** per-class thresholds are fitted on the OOF predictions of the other 4 folds only, then probabilities are rescaled so the threshold maps to 0.5. This avoids tuning on the fold being scored.
- **Uncertainty:** 95% lesion-bootstrap CIs; model comparisons use a paired bootstrap on the same lesions.
- **Decision rules** were stated before running where possible (e.g. augmentation adopted only if late fusion gains ⩾ +0.01).

3.2 Models

Step 1 — non-image floor. Logistic regression and LightGBM on demographics and MONET scores, to set the score an image model must beat.

Step 2 — frozen linear probe. Four backbones (PanDerm-B, DINOv2-B, EVA-02-B, EVA-02-S) with frozen 224 px embeddings and a class-balanced softmax regression. Purpose: pick backbones cheaply and test crop vs pad, and whether the clinical image and metadata add signal.

Step 3 — full fine-tuning, dermoscopy only. PanDerm-B, DINOv2-B and EVA-02-B. Setup: AdamW, layer-wise lr decay 0.75, 2 warm-up epochs, cosine lr 1e-4 → 1e-6, 25 epochs, batch 64, drop-path 0.1, bf16, horizontal-flip TTA. Imbalance handled with logit-adjusted

cross-entropy ($\tau = 1$). Augmentation: random resized crop (scale 0.35–1.0), flips, 90° rotations, mild colour jitter. No early stopping or checkpoint selection: the last-epoch model is evaluated, so the held-out fold never influences training.

Step 4 — add the clinical image. Three designs: (a) *late fusion* — a separate model per view, predictions combined by geometric mean 50/50; (b) *shared* — one backbone for both views; (c) *sep* — two backbones trained jointly. Late fusion was then run with 3 seeds.

Steps 5, 8–10 and side experiments. Each tested one idea on top of the Step 4 late-fusion model: metadata fusion (5), EVA+PanDerm ensemble (8), classifier re-training with class balancing (9), epoch budget (10), plus an augmentation A/B, 448 px input (Step 7) and label-shift correction. Step 6 has no recorded experiment in the repository; the numbering is kept as used in the code.

3.3 Why these choices

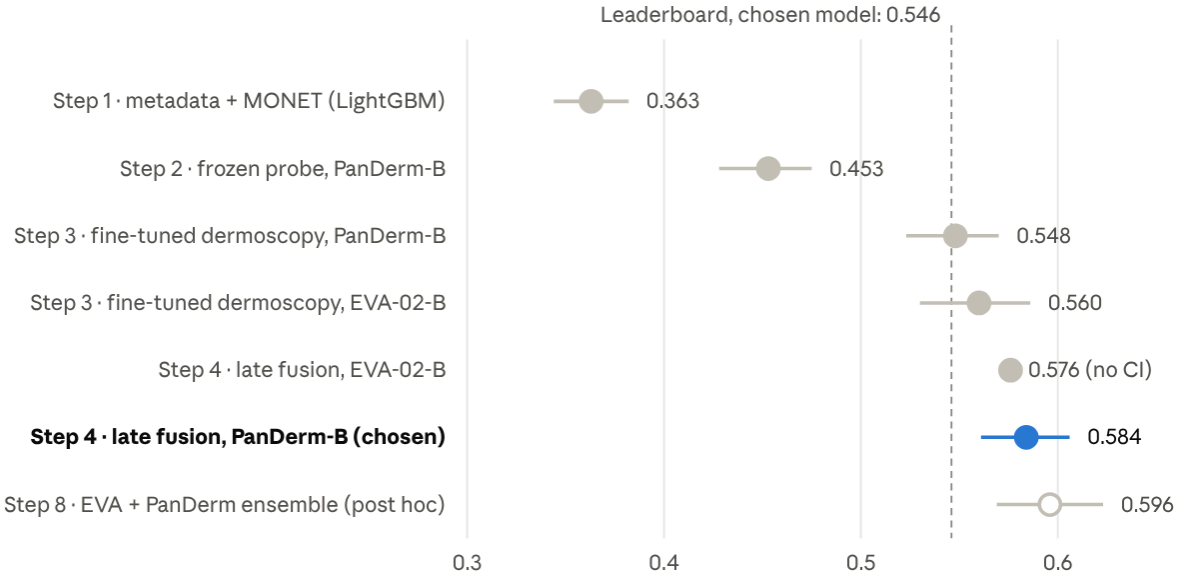
- **PanDerm-B and EVA-02-B** were carried forward because they led the probe and fine-tuning results; DINOv2-B was dropped after Step 3 (0.531 at seed 0, below both).
- **Late fusion** was preferred because it can be trained and checked per view and it was at least as good as joint models.
- **Logit adjustment** was chosen because the metric weights all classes equally while training data are 48% BCC.
- **Threshold calibration** was necessary because the ≥ 0.5 rule otherwise under-predicts rare classes.

4 Experiments and results

CV macro F1 rose from 0.363 (metadata only) to 0.584 (PanDerm-B late fusion); the leaderboard score of that model is 0.546.

CV macro F1 rose from 0.363 to 0.584 through fine-tuning and the clinical view

Calibrated macro F1, 5-fold CV, 95% lesion-bootstrap CI; Step 3 onwards = 3-seed ensembles



Hollow point: combination chosen after seeing OOF results (optimistic). Dashed line: test-set score of the chosen model.

results/step1–8 READMEs and summary files, repo 260925MILK10k

The EVA-02-B and PanDerm-B late-fusion models do not differ resolvably (EVA − PanDerm −0.008, −0.029..+0.014). PanDerm-B was submitted because its gain from the clinical view was the only one with a CI above zero. This is a weak criterion: a CI that excludes zero for one model and not the other does not show that the models differ, so the choice between the two is effectively open.

Leaderboard submissions (479 test lesions):

Date	Model	CV macro F1 (95% CI)	Leaderboard	Rank
2026-09-29	PanDerm-B late fusion, 448 px, seed 0	0.592 (0.563–0.622)	0.545	–
2026-09-28	EVA + PanDerm late-fusion ensemble (chosen post hoc)	0.596 (0.569–0.623)	0.505	–
2026-09-28	PanDerm-B late fusion, 224 px, 3 seeds	0.584 (0.561–0.606)	0.546	16

The leaderboard penalised the post-hoc ensemble that CV had flagged as optimistic. The 448 px run matched the 224 px ensemble with one seed, but the paired CV comparison was not resolved.

5 What worked

Five measures produced a gain whose 95% CI excludes zero, or a consistent gain across seeds. All deltas are paired bootstrap on CV, calibrated macro F1.

Measure	Why it was tried	Result
Image embeddings instead of metadata only	Metadata and MONET scores carry limited diagnostic signal	Frozen PanDerm-B probe 0.453 vs LightGBM 0.363: +0.090 (+0.062..+0.114)
Centre crop instead of padding (probe)	Padding wastes input pixels on 600×450 images	PanDerm-B dermoscopy +0.024 (+0.008..+0.039)
Full fine-tuning instead of frozen features	Frozen features are not adapted to 11 dermatology classes	EVA-02-B seed 0 +0.150 vs its frozen dermoscopy probe; PanDerm-B seed 0 +0.092 vs best frozen probe
Clinical image by late fusion	The clinical view shows context the dermoscope misses	PanDerm-B 3-seed: 0.548 → 0.584, +0.036 (+0.018..+0.054); EVA-02-B +0.016 (−0.007..+0.039)
Nested per-class threshold calibration	The ⩾ 0.5 rule under-predicts rare classes	e.g. LightGBM raw 0.304 → 0.363; EVA-02-B seed 0 raw 0.539 → 0.551

Two supporting findings:

- **The late-fusion gain comes from the second view, not from a second network.** Fusing two dermoscopy models of different seeds gave 0.542–0.559, no better than one. The largest per-class gains from the clinical view were in INF, BEN_OTH, BKL, AKIEC, DF and SCCKA.
- **Seed ensembling helps a little and reduces variance.** Single-seed late-fusion PanDerm-B scored 0.578 / 0.603 / 0.578; the 3-seed ensemble 0.584. The seed range (0.025) is as large as most effects tested later, which is why single-seed comparisons were treated with caution.

6 What did not prove useful

"No gain" in this section means no gain resolvable at this sample size (9 MAL_OTH lesions in CV; seed-to-seed range ~0.025); small true effects cannot be excluded, and no correction for multiple comparisons was applied. None of the following beat the late-

fusion PanDerm-B model by a resolved margin; stacking and a plain classifier refit made it measurably worse. Deltas are paired bootstrap, 95% CI, calibrated macro F1, CV unless marked LB (leaderboard).

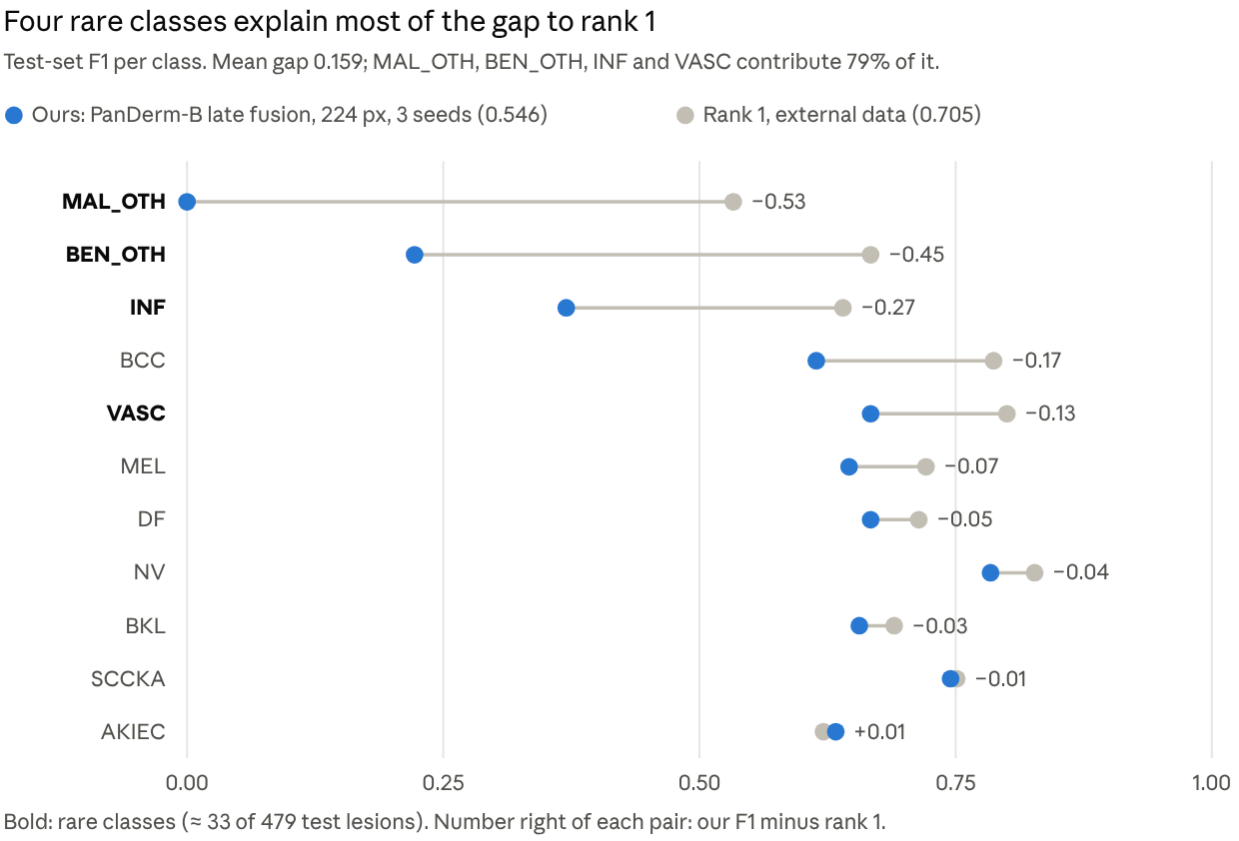
Idea	Why it was tried	Result	Verdict
Stacking (meta-learner) for view fusion	Learn fusion weights instead of fixing 50/50	EVA −0.047 (−0.072..−0.022); PanDerm −0.034 (−0.054..−0.014)	Worse; geometric mean kept
Joint two-view models (shared / separate backbones)	One model could learn cross-view features	PanDerm seed 0: shared 0.565, sep 0.572 vs late 0.578; EVA late − sep +0.018 (+0.000..+0.036)	Not better; late fusion kept
DINOv2-B fine-tuned	Strong general backbone, near-best in the probe	0.531; EVA − DINOv2 +0.020 (−0.010..+0.049)	Not carried forward
Metadata / MONET late-fused with images	Probe showed +0.010 (+0.002..+0.018) from metadata	Nested weight choice: −0.001 to +0.011, all n.s.; 50/50 weighting −0.024 to −0.063	No robust gain
HiV15 augmentation pipeline (from the melanoma project)	Worked in a related binary task	With grayscale: late −0.018 (−0.035..+0.000), clinical −0.046; without: late −0.005	Original augmentation kept
448 px input (PanDerm-B, seed 0)	Native images are 600×450; 224 px loses detail	Late +0.014 (−0.008..+0.036), gain almost only MAL_OTH (1 more TP of 9); AUC slightly lower; LB 0.545 vs 0.546	Within seed noise
Label-shift correction (BCTS + EM prior + plug-in F1)	Test prior differs strongly from training	Simulated n = 479 sets: EM plug-in −0.037 to +0.005; knowing the true prior would give +0.008 to +0.079	EM too noisy at n = 479; not submitted

Idea	Why it was tried	Result	Verdict
Classifier re-training with class balancing (cRT, τ-norm)	Standard long-tail fix	Plain refit −0.018 (−0.033..−0.003); balanced refits −0.006; τ-norm +0.008 (−0.002..+0.018); refits lose ~0.015 mean AUC	Frozen features are nearly memorised; no gain
Epoch budget 15–50 (default 25)	Check under/over-training	0.569–0.584, all n.s. vs 25; clinical model degrades beyond 25; averaging last epochs −0.008..+0.006	20–30 is a flat optimum; 25 kept
EVA + PanDerm late-fusion ensemble	Two different backbones may complement	CV 0.596 (combination chosen after seeing OOF, so optimistic); LB 0.505 vs 0.546	Rejected; changed rare-class decisions

Common pattern: most ideas changed the **ranking** of lesions very little (mean AUC stayed around 0.93–0.94) and moved F1 mainly through a handful of rare-class lesions near the threshold. With 9 MAL_OTH lesions in CV and ~10 in test, such moves are within noise.

7 Discussion and limitations

Where the score is lost. On the test set the model ranks well (mean AUC 0.929 vs the leader's 0.905) and is on par with rank 1 on the six frequent classes. The gap of 0.159 macro F1 comes 79% from four rare classes — MAL_OTH, BEN_OTH, INF, VASC — about 33 of 479 test lesions. MAL_OTH is 0.00 F1 in every submission. Rank 1 declared external data and has much higher average precision on these classes (e.g. INF 0.61 vs 0.26, BEN_OTH 0.52 vs 0.22 in the 448 px run). Their method is not known beyond the name "3D-ViT v3", so their advantage cannot be attributed to external data alone.



ISIC leaderboard details pages (submissions/leaderboard/*.md), 2026-09-28

Two likely reasons — hypotheses consistent with the data, not proven.

1. *Prior shift.* BCC is 48% of training but 13% of test. Thresholds fitted on training OOF therefore over-call BCC on test: BCC F1 falls from 0.91 in CV to 0.61 on the leaderboard. Correcting for it without labels (EM) was too noisy at n = 479.
2. *Missing rare-class examples.* MAL_OTH has 9 training lesions, all collision tumours, while the allowed test diagnoses include Kaposi sarcoma, Merkel cell carcinoma, atypical fibroxanthoma, T/NK lymphoma and MPNST — none in training. BEN_OTH, INF and VASC also have allowed test diagnoses absent from training (e.g. scar, cylindroma, verruca, lymphangioma). Which of these actually occur in the test set is not disclosed.

Re-weighting the same few rare examples (cRT, balanced losses) did not help, which is consistent with a lack of examples rather than a lack of weight.

Threats to validity.

- **Statistical power.** Many side experiments ran with seed 0 only; the seed range (~0.025) is larger than most effects tested. Negative results mean "not resolvable here", not "no effect". About 15 comparisons were made without multiple-comparison correction.

- **CV vs leaderboard.** CV and leaderboard differ by ~0.04 (0.584 vs 0.546). Prior shift is the likely but unproven cause. CV rankings of small effects did not always hold on the leaderboard (EVA+PanDerm: CV +0.012, LB −0.041).
- **Leaderboard reuse.** Three submissions were made, and per-class test counts were derived from the feedback. These were not used for tuning, but they shaped the analysis in Section 7.
- **Possible leakage, size unknown.** No patient ID exists, so folds may place lesions of the same patient in training and validation. PanDerm-B pretraining data may overlap with MILK10k sources (not verified). Both would inflate CV.
- **Missing comparisons.** No comparison with the baselines reported in the MILK10k paper; no per-class CV table or calibration plot; the hierarchical (granular) labels were not used; alternative losses implemented in the code (weighted CE, focal) are not reported.
- **Not tested yet:** a hierarchical auxiliary head on the 48 granular diagnoses, and a metadata branch trained jointly with the image model.

8 Conclusion and next steps

With MILK10k data alone, the best configuration is PanDerm-B fine-tuned per view with late fusion and calibrated thresholds: 0.584 CV, 0.546 leaderboard. The large, resolved gains came from the basics — fine-tuning, the second image view and threshold calibration. Refinements of training, input size, balancing, metadata and ensembling did not move the score beyond noise.

The evidence suggests one lever: more and more varied rare-class examples, in both modalities. Planned next steps:

- [ ] Add external rare-class images (MAL_OTH, BEN_OTH, INF, VASC) to training folds only, after checking for duplicates with the MILK10k test set, licences per source, clinical-image availability, and label mapping via ISIC-DX.
- [ ] Exclude the MILK10k images that are already in the ISIC archive ("MILK study team", 5,211 lesions).
- [ ] Keep evaluation on the MILK10k folds; compare against 0.584 CV / 0.546 LB.
- [ ] Test a hierarchical auxiliary head on the granular diagnoses.

References and project sources

- MILK10k Benchmark challenge description (project doc MILKChallenge.rtf); organisers H. Kittler, P. Tschandl, Medical University of Vienna.

- MILK10k paper (project file MILK10k--A-Hierarchical-Multimodal-Imaging-Learnin.pdf; title truncated in the file name).
- Kim C, Gadgil SU, DeGrave AJ, et al. Transparent medical image AI via an image-text foundation model grounded in medical literature. *Nat Med.* 2024;30(4):1154–1165. doi:10.1038/s41591-024-02887-x (MONET).
- Menon AK et al. Long-tail learning via logit adjustment. ICLR 2021.
- Project notes: claude/MILK10k_learnings_2026-09-28.md, claude/MILK10k_external_data_inventory_2026-09-28.md.
- Repository 260925MILK10k: results/step1–step10 READMEs and summaries, results/priorshift/README.md, submissions/LOG.md, submissions/leaderboard/*.md.